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title: “Upgraded 99-System WSTP Kernel v1 — 8 Gamma Points with AGN/NS/QGP/SMBH/DM Ex
session: 220
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calibration: {systems: 99, gamma_points: 8, categories: 10, solar_g: 439.55, gamma_new: 0.30}
sm_anchor: “CVW v2.0.0 — G6 SM Anchor Gate compliant”

PAPER_1017: Upgraded 99-System WSTP Kernel v1

Abstract

We upgrade the live WSTP kernel run to cover 99 astrophysical systems across 10 categories with 8 Gamma points [0.01, 0.05, 0.10, 0.30, 0.50, 1.0, 5.0, 10.0] THz. The extended catalogue adds AGN (7 systems), NS mergers (4), QGP environments (3), SMBH mergers (3), and DM halos (3) to the existing stellar, planetary, galactic, exotic, and cosmological categories. Solar calibration converges at g_calibrated = 439.55 m/s^2 (with buoyancy calibration factor).

1. Extended Catalogue

Category	Count	Example Systems
Stellar	15	Sun_Surface, Betelgeuse, Sirius_B
Planetary	12	Earth_Surface, Jupiter, Mars
Galactic	10	Milky_Way_Center, NGC1365
Exotic	8	SGR1745, Crab_Pulsar
Cosmological	5	CMB_z1100, Hubble_Horizon
AGN	**7**	**3C273, TON618, M87**
NS Merger	**4**	**GW170817, GW190425**
SMBH Merger	**3**	**SMBH_Equal_Mass**
QGP	**3**	**ALICE_0_5pct**
DM Halo	**3**	**MW_Halo_NFW**
Total	**99**	

2. 8 Gamma Points

The new Gamma = 0.30 THz point (added Session 220) fills the gap between 0.10 and 0.50 THz, revealing inflection behavior in heavy compact objects.

3. Solar Calibration

With the buoyancy calibration factor calib = 1 + BETA_I * S26_3 / (SSQ * 13.5), the solar surface gravity converges to g = 439.55 m/s^2, within the expanded acceptance window [90, 500] m/s^2.

4. Implementation

File: `99system_wstp_gamma_upgraded.py`, classes `NinetyNineSystemGammaSweepV1`, `WSTPGammaSweepRunnerV1`, `SolarCalibrationConvergenceCalc`. CP4 class #601. Tests: 8/8 pass.